Factoring trinomials

1 Determine whether or not the following expressions are in factored form.

a $-5(2y+z)-b(2y+z)$

b $(a+7)(a+2)$

c $(x-5)(x+y)$

d $3x(a+2b)+2(a+2b)$

2 Factor the following expressions by taking out the common factor.

a $5(a+b)+v(a+b)$

b $x(y-z)-w(y-z)$

c $8x(2y+3w)-z(2y+3w)$

d $8z\left(5x^2+4y\right)-\left(5x^2+4y\right)$

e $8y(y-4)+10(4-y)$

f $(4c-d)(c+9d)-5(d-4c)$

3 Factor the following expressions by grouping in pairs.

a $x^2+5x+8x+40$

b $x^2-5x+10x-50$

4 Factor completely the following expressions:

a $xy-9x+2y-18$

b $56-8y-7x+xy$

c $12xy+wx+12yz+wz$

d $2x+xz-40y-20yz$

e $50+5y+10x+xy$

f $12xy+wx+12yz+wz$

5 By rewriting $4x^2+17x+4$ as an expression having four terms and factoring in pairs, factor the expression completely.

6 Complete the following sentence:
To factor the quadratic expression $x^2+11x+24$, we need to find two numbers whose product is $\square$ and whose sum is $\square$.

7 For the following, find the values of a and b given $a<b$ and:

a $a+b=15$ and $a\times b=56$

b $a+b=-15$ and $a\times b=50$

8 For the following, find the values of a and b given that $b<a$ and:

a $a+b=-11$ and $ab=18$

b $a+b=-6$ and $ab=-112$.

9 Complete the expressions that will make the following equations true.

a $(m+8)(\square) = m^2 + 18m + 80$

b $y^2 - 8y - 65 = (y-13)(\square)$

10 Consider the given diagram, where a large rectangle is made up of four smaller rectangles:
One expression for the area of the rectangle is $m^2 + 14m + 45$. Find the area of the large rectangle in factored form.

	m	9
5	$5m$	45
m	m^2	$9m$

11 Consider the given diagrams.
Find an expression for the total area of the rectangle in factored form.

a

x^2	$5x$
$4x$	20

b

$3x^2$	x
$18x$	6

12 Factor the following trinomials:

a $x^2 + 6x + 5$

b $x^2 + 14x + 40$

c $x^2 - 5x + 4$

d $x^2 - 13x + 30$

e $x^2 - 15x + 56$

f $x^2 + x - 132$

g $x^2 + 9x - 90$

h $x^2 - 2x - 8$

i $x^2 - 3x - 70$

j $6 + 5x + x^2$

k $44 - 15x + x^2$

l $-12 - 13x - x^2$

m $-9 + 10x - x^2$

13 Factor the following expressions completely[redacted]rst taking out a common factor:

a $4x^2 + 56x + 192$

b $-5x^2 + 10x + 40$

c $3x^2 - 21x - 54$

d $2x^3 + 16x^2 + 30x$

14 A ball is thrown from the top of a 140 m tall cliff, with an initial velocity of 50 m/s. The height of the ball after t seconds is very well approximated by the quadratic expression $-10t^2 + 50t + 140$. Fully factor this quadratic expression.

15 Tara is practising diving. Tara springs up off a board 32 feet high, and after t seconds, her height in feet above the water is described by the polynomial $-16t^2 + 16t + 32$.

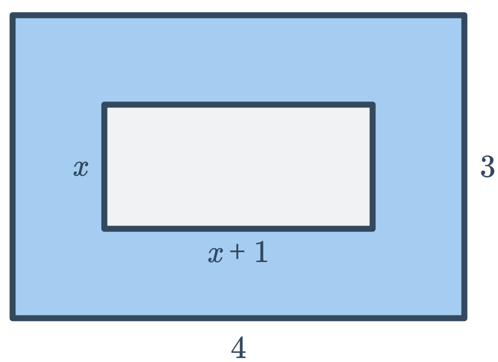
a Factor completely the polynomial.

b Substitute $t = 2$ into the factored polynomial and find the value of the expression.

c Substitute $t = 2$ into the original polynomial and check the value of the expression.

d Explain the implications of part (b) and (c) of this question.

16 Consider the given diagram:
Write down an expression in factored form for the shaded area in the diagram.



17 Complete the following equations to show the correct factors of the given expressions.

a $3x^2 + 10x - 8 = (3x - 2)(x + \square)$

b $12x^2 + 25x + 7 = (4x + 7)(3x + \square)$

c $15x^2 - 13x - 6 = (5x - \square)(3x + \square)$

18 Factor following trinomials:

a $7x^2 - 75x + 50$

b $6x^2 + x - 7$

c $9x^2 - 19x + 10$

d $7x^2 - 34x - 48$

e $6x^2 + 13x + 6$

f $-12x^2 - 7x + 12$

g $90 - 89x + 8x^2$

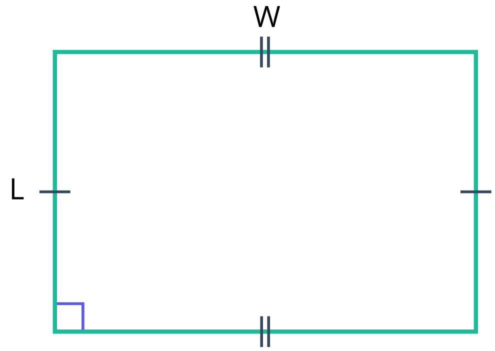
h $8 - 14p - 49p^2$

19 A rectangle has an area of $6x^2 + 23x + 20$. If the length and width are linear factors of $6x^2 + 23x + 20$, find the dimensions of the rectangle.

20 Consider the given rectangle:

a. Let the length $L = -56y + 11$ and the width $W = 5y^2$.
Write the perimeter of the rectangle in terms of y. Express the perimeter in simplified expanded form.

b. Factor the polynomial completely.



21 Find all positive and negative integers k such that $x^2 + kx + 14$ is factorable.

22 Find all positive integers c such that $x^2 + 4x + c$ can be factored into the product of linear factors that contain integer terms.

Additional questions

23 A square carpet with side length x is modified by shortening one side by 6 inches and lengthening the other by 7 inches.

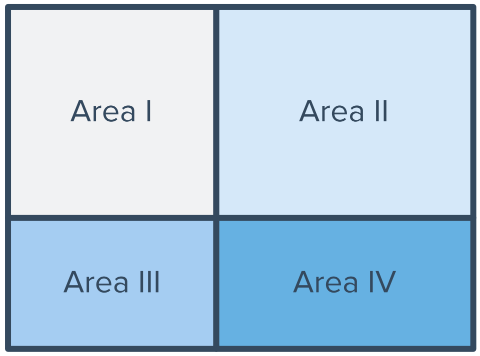
a. Find the area of the carpet in the form $(x + p)(x + q)$.

b. Find the area of the modified carpet in the form $x^2 + bx + c$.

c. Describe the relationship between the values 7, -6, 1 and -42.

24 The area of a quilt can be expressed as $x^2 +$ [illegible] $4x + 24$.

a Label the area model so that it represents the area of the quilt.



b Express the area of the quilt in factored form $(x + p)(x + q)$.

25 Suppose we want to write the trinomial $x^2 + bx + c$ in the form $(x + p)(x + q)$. Read each condition and explain what must be true regarding the signs of p and q.

a c is negative.

b c and b are both positive.

c c is positive and b is negative.

26 Find two digits $(1$ to $9)$ to fill in the blanks and create a trinomial in the form $ax^2 + bx + c$ that

a Maximizes the value of b.
$(x + \square)(x + \square)$

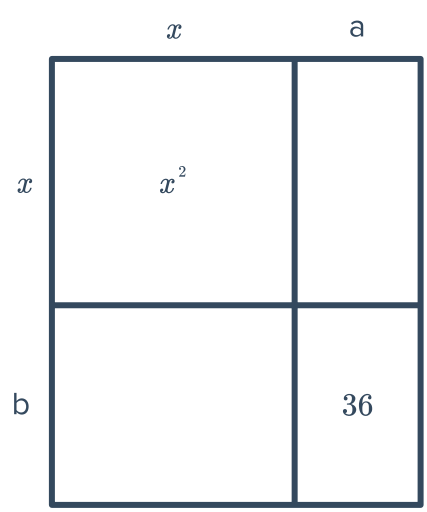
b Minimizes the value of c.
$(x + \square)(x + \square)$

c Minimizes the value of b.
$(x + \square)(x - \square)$

27 A square field with side length x will have its length extended by a and its width extended by b.

a What values of a and b would maximize the length of the entire field?

b What values of a and b would minimize the difference in area of the bottom left and top right field?



28 Think about factoring the quadratic expression $15x^2 + 14x - 16$. In the factored form $(Ax + C)(Bx + D)$, complete the following statements so that they are true.

a A and B multiply to be ▯.

b The signs for A and B will be ▯.

c C and D multiply to be ▯.

d The signs for C and D will be ▯.

29 Think about factoring the quadratic expression $-45x^2 + 29x - 4$. In the factored form $(Ax + C)(Bx + D)$, complete the following statements so that they are true.

a A and B multiply to be ▯.

b The signs for A and B will be ▯.

c C and D multiply to be ▯.

d The signs for C and D will be ▯.

30 Brandon claims that the polynomial $3x^2 + 15x - 42$ will follow the factoring form of trinomials with a leading coefficient $a \neq 1$.
Explain Brandon's error, then fully factor the expression.

31 Using the digits 1 to 9 with no repeats, fill in the blanks to create a factorable trinomial and provide the factored form.

a $4x^2 + \square\square x + \square$

32 Consider each of the following polynomials

i $2x^2 + x + 1$

ii $3x^2 - 2x - 4$

iii $6x^2 + 10x - 5$

iv $12x^2 - 12x + 1$

a What do the polynomials have in common?

b How do you determine if a polynomial where $a \neq 1$ is not factorable?

33 A photographer wants to put a border around her photo.

a Why might the photographer use x to represent the width of the border?

b Use factoring to find expressions for the dimensions of the photograph with its border if the total area is $4x^2 + 24x + 35$.

c What is the area of the photograph? Explain how you reached your solution.

34 Describe the similarities and differences between factoring a polynomial in the form $ax^2 + bx + c$ where $a = 1$ or where $a \neq 1$.

35 Consider the factorable polynomial $12x^2 +$ [redacted], where b is a positive integer.

a Find the largest possible value for b and list the factors.

b Find the smallest possible value for b and list the factors.